INDIA ELECTRONICS MANUFACTURERS E-WASTE COMPLIANCE & GAP ANALYSIS REPORT

Review of the 10 manufacturers shown in the supplied reference image
FY 2025–26 context | Public disclosures reviewed through 01 September 2026

Regulatory basis	E-Waste (Management) Rules, 2022 and amendments; EPR requirements; RoHS; Battery Waste Management Rules, 2022 and Plastic Waste Management Rules, 2016 where separately applicable.
Purpose	Identify documented practices, public-evidence gaps and verification priorities. This is a desk review, not a finding of legal violation.
Company set	Foxconn India, Dixon Technologies, Elin Electronics, VVDN Technologies, Syrma SGS, Tata Electronics, Avalon Technologies, Centum Electronics, Sahasra Electronics and Kaynes Technology.

Prepared for academic / analytical use

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Critical interpretation: “Not publicly demonstrated” means the reviewed evidence did not establish the requirement. It does not mean the company is violating the law.

1. Executive Summary

The E-Waste (Management) Rules, 2022 form India's principal e-waste framework. CPCB states that applicable manufacturers, producers, recyclers, refurbishers and dismantlers are covered; Producers of covered EEE have EPR obligations and use the CPCB portal and registered recyclers for fulfilment. [R11][R12]

The ten-company review shows strong operational evidence at several firms—especially authorised recycler arrangements, collection points, EPR authorisation and waste-management controls. The most common weakness is **visibility of current regulatory evidence**: CPCB EPR registration numbers, exact targets, EPR certificate fulfilment and quarterly/annual return records are not consistently disclosed.

No confirmed e-waste-law violation is asserted for any company. A disclosure gap is not a legal violation; final determination requires CPCB/SPCB records, inspections, notices, registration data and returns.

Finding	Assessment
Strong public e-waste evidence	Sahasra, Syrma SGS, Dixon, Kaynes
Moderate / incomplete evidence	Elin, VVDN, Tata Electronics, Avalon, Centum
Limited Indian-specific evidence	Foxconn India
Confirmed violations established	None from reviewed evidence
Main system-wide gap	Current EPR/return/target evidence and role classification

The report follows the structure of the supplied Kerala State E-Waste Management Report: executive summary, regulatory context, methodology, compliance review, gap analysis, scorecard, recommendations and annexures.

2. Introduction & Regulatory Context

2.1 Regulatory framework

The E-Waste (Management) Rules, 2022 took effect from 1 April 2023 and replaced the 2016 framework. CPCB identifies EPR, stakeholder registration, recycling through registered recyclers, quarterly and annual returns, audits and environmental compensation as key features. [R11][R12]

CPCB's current E-Waste EPR Portal states that applicable Producers obtain registration certificates and EPR targets/obligations through the portal and reminds registered Producers to meet applicable targets and file quarterly returns. [R11]

2.2 Requirements used in this report

Requirement	Compliance question
E-Waste Rules 2022 registration	Is the entity a covered Manufacturer/Producer/Recycler/Refurbisher, and is current registration evidenced?
EPR obligation	If the company is a Producer, is the target and fulfilment evidenced through registered recyclers/EPR certificates?
Quarterly / annual returns	Is current reporting evidence publicly demonstrated?
Authorised recycler channelisation	Is e-waste transferred through registered/authorised channels?
RoHS	Are hazardous-substance restrictions evidenced for covered EEE?
Records / auditability	Can waste quantities, transfers, recycling and certificates be traced?
Environmental compensation	Has any official violation/EC order been established?
Separate waste EPR	Battery and plastic obligations are assessed separately where applicable

RoHS note: CPCB's FAQ describes restrictions for lead, mercury, cadmium, hexavalent chromium, PBB and PBDE in covered EEE. [R12]

3. Methodology & Evidence Classification

3.1 Scope

The company universe is taken from the supplied image. The original source page corresponding to that image lists the ten companies as Foxconn India, Dixon, Elin, VVDN, Syrma SGS, Tata Electronics, Avalon, Centum, Sahasra and Kaynes. [R13]

3.2 Evidence classification

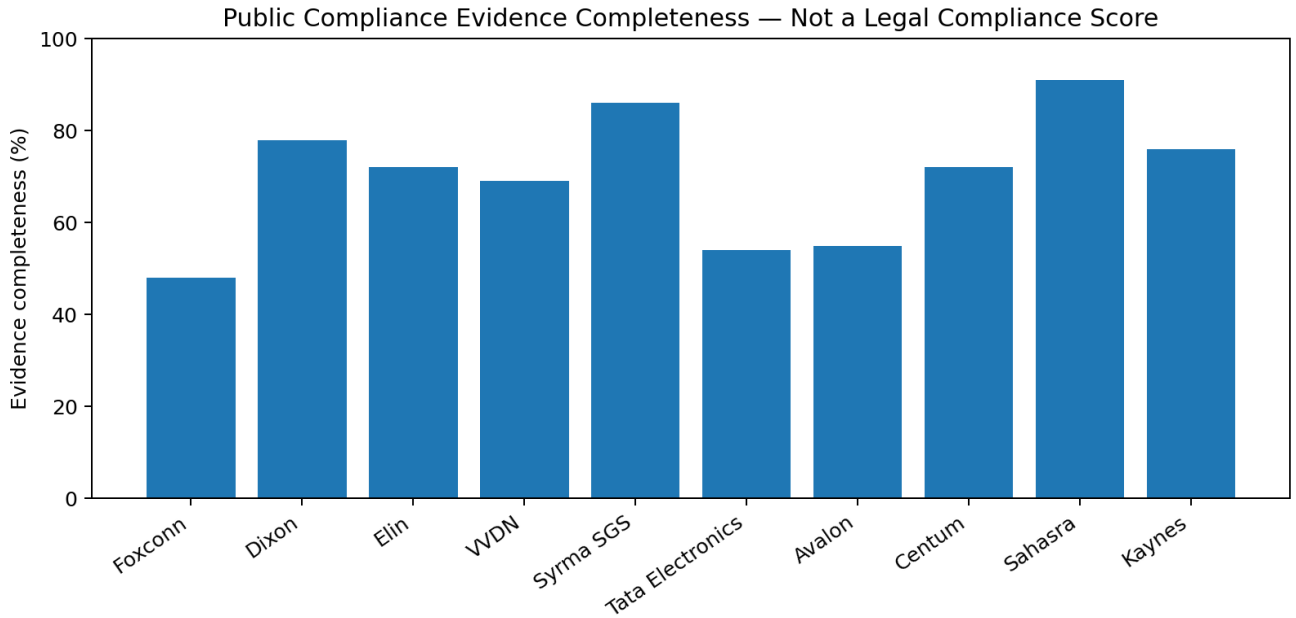
Class	Definition
Documented	Explicit company or government evidence establishes the practice.
Partially documented	Related practice is disclosed, but the exact current legal obligation is not established.
Not publicly demonstrated	No reliable current evidence was located in the reviewed material.
Role-dependent	The requirement depends on whether the entity legally qualifies as a Producer or another covered role.
Unverified	Indian applicability/entity/status could not be established from the reviewed evidence.

3.3 Important limitation

Annual reports and corporate webpages are not substitutes for CPCB registration databases, inspections or legal orders. In particular, B2B EMS companies may generate manufacturing-process e-waste without being the Producer responsible for end-of-life EEE under the EPR definition. Therefore, the report does not infer a violation merely because a company does not publish an EPR certificate number.

3.4 Evidence-completeness indicator

The comparative score below is only a visual indicator of how complete the public evidence appears. It is not a legal compliance score.



4. Company Compliance Review

“Gap” below means a public-evidence gap, not a confirmed violation.

Company	Evidence	Documented practices	Main gap
Foxconn India	Limited public evidence	Government record for relevant used-EEE imports requires end-of-life disposal under E-Waste Rules 2022 + HOWM Rules [R10]	Current public EPR registration/target/recycler evidence not found [R10]
Dixon Technologies	Strong	Authorised recyclers, 3R waste system and RoHS controls; B2B end-of-life waste handling [R1]	Current 2022 EPR registration/target not clearly demonstrated [R1]
Elin Electronics	Moderate–Strong	E-waste management COP disclosed historically; FY2024–25 report says waste COP under E-Waste Rules 2022 [R2]	Current waste COP under EPR and Plastic Rules 2022 registration details need [R2]
VVDN Technologies	Moderate	EPR plan and recycler/dismantler arrangement with collection channel [R3]	Public waste page still cites 2016/2018 rules; current 2022 evidence shou [R3]
Syrma SGS	Strong	Authorised recycler, collection points and EPR activities; B2B end-of-life treatment disclosed [R4]	Exact current EPR target/return evidence not publicly established [R4]
Tata Electronics	Moderate	MoEFCC record specifies E-Waste Rules 2022 + HOWM requirements for relevant used-EEE import [R5]	Current company-level e-waste EPR registration/target/returns not establish [R5]
Avalon Technologies	Moderate	FY2024–25 filing reports 2.95 t e-waste and waste recovery/disposal data [R6]	Dedicated EPR registration/target/recycler evidence not clearly demo [R6]
Centum Electronics	Moderate–Strong	FY2024–25 report quantifies 6.30 t e-waste and waste recovery/disposal data [R7]	Disposal/EPR status role-dependent; current product-scope/EPR eviden [R7]
Sahasra Electronics	Strong	EPR authorisation, e-waste authorisation, collection/reverse logistics and recycler details [R8]	Dedicated e-waste webpage still cites 2016/2018 terminology; current 2022 [R8]
Kaynes Technology	Strong	E-waste, plastic and biomedical waste handed to authorised agencies; current EPR registration/target and procurement [R9]	Current EPR registration management and quantified return evidence not independe [R9]

The strongest evidence is found where companies publish dedicated EPR/e-waste policies or detailed annual-report disclosures. Sahasra, Syrma SGS, Dixon and Kaynes are particularly clear on operational waste handling; Elin also provides current EPR-related disclosure. [R1][R2][R4][R8][R9]

5. Company-by-Company Findings — Part I

5.1 Foxconn India

The reviewed government evidence is strongest for specific used-EEE import conditions rather than a general corporate EPR profile. A MoEFCC record requires end-of-life disposal of relevant imported used EEE under the prevailing HOWM Rules and E-Waste Rules, 2022. [R10]

Public-evidence gap: Current Indian EPR registration/target and routine e-waste reporting not established in reviewed sources

5.2 Dixon Technologies

Dixon reports source segregation, the 3Rs, government-authorised recyclers and RoHS controls. It also explains that its B2B model means it does not directly receive products at end of life. [R1]

Public-evidence gap: Current E-Waste Rules 2022 EPR registration/target not clearly demonstrated

5.3 Elin Electronics

Elin's FY2024–25 annual report states that it is registered under EPR and plastic-waste management rules. Earlier company material also disclosed a Certificate of Permission for e-waste management, with validity stated through 31 March 2025. [R2]

Public-evidence gap: Current e-waste COP / current E-Waste Rules 2022 registration details need verification

5.4 VVDN Technologies

VVDN's official e-waste page describes an EPR plan and a recycler/dismantler arrangement for collection and recycling. The public page still cites the 2016/2018 framework, which should be updated to the current 2022 regime. [R3]

Public-evidence gap: Public e-waste page still cites 2016/2018 rules; current 2022 evidence should be updated

5. Company-by-Company Findings — Part II

5.5 Syrma SGS

Syrma SGS states that an authorised recycler handles collection, transport, storage, segregation, dismantling and recycling, with collection points. Its FY2024–25 report also explains that it is a B2B EMS provider and does not control the end-of-life handling of finished goods sold by OEM customers. [R4]

Public-evidence gap: Exact current numerical EPR target/return evidence not publicly established

5.6 Tata Electronics

A MoEFCC Expert Committee record for Tata Electronics Systems Solutions specifies that relevant used EEE imported for manufacturing must be disposed of at end of life under the E-Waste Rules 2022 and HOWM Rules. [R5]

Public-evidence gap: Current company-level e-waste EPR registration/target/returns not established

5.7 Avalon Technologies

Avalon's FY2024–25 filing reports 2.95 tonnes of e-waste generated and provides waste recovery/disposal data. The review did not establish current e-waste EPR registration/target evidence. [R6]

Public-evidence gap: Current e-waste EPR registration/target/recycler evidence not clearly demonstrated

5. Company-by-Company Findings — Part III

5.8 Centum Electronics

Centum's FY2024–25 report quantifies 6.30 tonnes of e-waste and provides waste recovery/disposal data. Whether Producer EPR applies is activity- and product-scope dependent; it should be documented rather than assumed. [R7]

Public-evidence gap: Producer EPR status is role-dependent; current product-scope/EPR evidence needs verification

5.9 Sahasra Electronics

Sahasra publishes EPR authorisation and e-waste management authorisation, and describes PAN-India collection, reverse logistics and an authorised recycler/dismantler. Its sustainability material also states EPR registration and RoHS/REACH controls. [R8]

Public-evidence gap: Dedicated e-waste webpage still uses 2016/2018 terminology; current 2022 details should be updated

5.10 Kaynes Technology

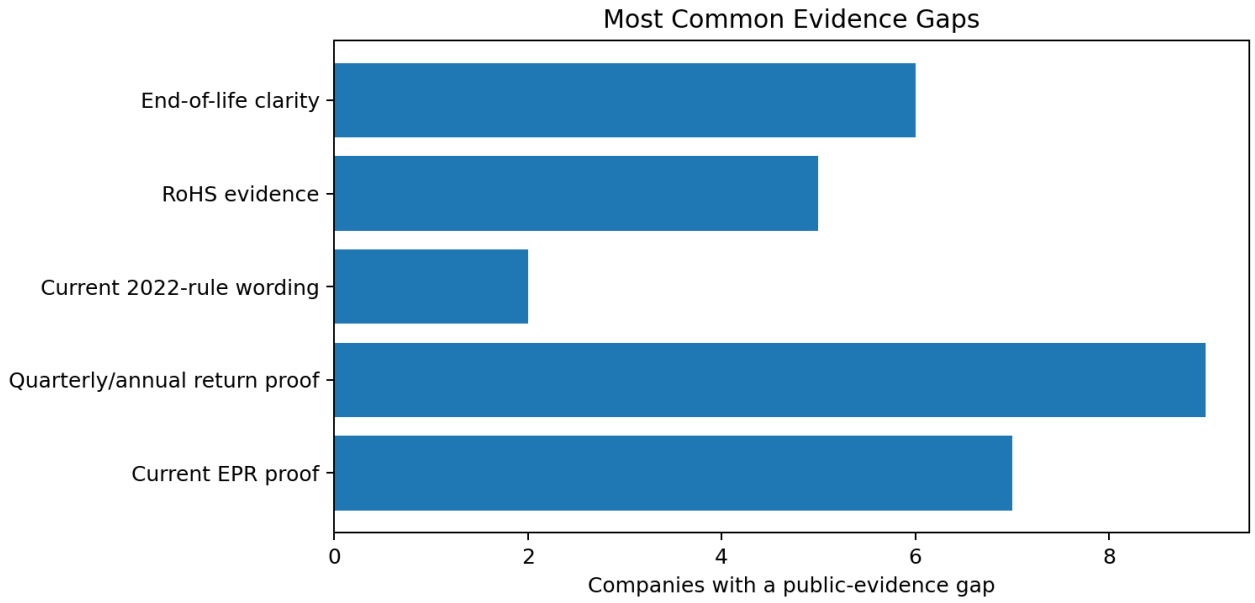
Kaynes reports that e-waste, plastic and biomedical waste are managed and handed to authorised agencies, supported by an environmental-management approach. [R9]

Public-evidence gap: Current EPR registration/target and quarterly-return evidence not independently established

6. Rules / Requirements Not Publicly Demonstrated

This section replaces the unsafe wording “rules not being followed” with the legally defensible formulation “requirements not publicly demonstrated.”

Requirement	Companies where evidence is incomplete	Interpretation
Current E-Waste Rules 2022 EPR registration	Foxconn, Dixon, Elin (current e-waste status), VVDN, Tata, Avalon, CPCB registration, before making a non-compliance claim	
Current EPR target / certificate fulfilment	Most companies in the reviewed public material	Annual-report narrative is not the same as CPCB target/certificate e
Quarterly / annual return evidence	Almost all companies	Public disclosure gap; CPCB records are the appropriate verificatio
Current 2022-rule wording	Sahasra and VVDN	Outdated webpage language; not proof of violation
RoHS evidence	Several companies	May exist through customer specifications/supplier controls; produc
End-of-life take-back clarity	B2B EMS entities including Dixon and Syrma SGS	Role-dependent; B2B manufacturing does not automatically equal P



No confirmed violation is recorded here because no official CPCB/SPCB violation order, show-cause notice or environmental-compensation order was established for the ten companies in the reviewed evidence.

7. Comparative Evidence Scorecard

Public-evidence completeness only — not a legal compliance score.

Company	Score	Interpretation
Foxconn	48/100	Limited
Dixon	78/100	Moderate
Elin	72/100	Moderate
VVDN	69/100	Moderate
Syrma SGS	86/100	High
Tata Electronics	54/100	Limited
Avalon	55/100	Limited
Centum	72/100	Moderate
Sahasra	91/100	High
Kaynes	76/100	Moderate

Score dimensions used

Dimension	Weight
Current role / EPR evidence	20%
Authorised recycler/channelisation evidence	20%
RoHS / hazardous-substance evidence	15%
Reporting / auditability	15%
Collection / end-of-life clarity	10%
Other applicable waste-regime evidence	10%
Currentness / primary-source quality	10%

The score should be read as a research prioritisation tool. A lower score means more verification is needed, not that the company is legally non-compliant.

8. Key Findings & Gap Analysis

8.1 Strongest areas

• Authorised recycler/channelisation arrangements are the most consistently demonstrated control.

• Sahasra provides unusually explicit evidence of EPR authorisation, e-waste authorisation and collection/reverse logistics. [R8]

• Syrma SGS provides a detailed authorised-recycler and collection-point arrangement while clearly stating its B2B end-of-life limitation. [R4]

• Dixon documents government-authorised recyclers and RoHS controls. [R1]

• Elin's current FY2024–25 disclosure explicitly states EPR and plastic-waste registration. [R2]

8.2 Weakest areas

• Current CPCB EPR registration numbers, targets and certificate fulfilment are not consistently visible.

• Quarterly and annual return evidence is generally absent from public company disclosures, although CPCB treats timely returns as a requirement. [R11]

• Some dedicated e-waste webpages still cite the superseded 2016/2018 rules.

• B2B EMS companies need clearer separation between process e-waste and end-of-life EEE.

• Foxconn India and Tata Electronics require stronger India-specific public evidence before broad compliance conclusions are drawn.

8.3 Legal-risk interpretation

No confirmed e-waste-rule violation was established from the reviewed public evidence. The appropriate action for an evidence gap is CPCB/company verification, not an allegation.

9. Recommendations

1. Maintain a one-page E-Waste Compliance Register: legal role, CPCB registration number, covered EEE, annual target, EPR certificates and return status.

2. Update all e-waste webpages to the E-Waste Rules 2022 and clearly archive legacy 2016/2018 references.

3. Disclose authorised recycler/dismantler name, registration status and scope of the agreement.

4. Separate manufacturing-process e-waste from end-of-life products in ESG reporting, especially for B2B EMS companies.

5. Maintain traceability from generation → segregation → storage → authorised transport → registered recycler → recycling output/EPR certificate.

6. Publish current RoHS testing or supplier-control evidence where commercially appropriate.

7. Report battery and plastic EPR separately from e-waste EPR.

8. Use CPCB portal records as the final verification source before making legal compliance claims.

Priority actions

Company	Priority
Foxconn India	Verify India-specific Producer role/EPR registration and routine reporting evidence.
Dixon	Verify current E-Waste Rules 2022 EPR status and publish current evidence.
Elin	Verify current e-waste authorisation/CPCB registration after the historical COP period.
VVDN	Update public policy to the 2022 Rules and publish current registration evidence.
Syrma SGS	Keep B2B end-of-life limitation and authorised-recycler evidence explicit.
Tata Electronics	Publish current e-waste EPR role/target/return evidence where applicable.
Avalon	Add current e-waste EPR/recycler evidence to ESG disclosures.
Centum	Document product-scope basis for EPR applicability/non-applicability.
Sahasra	Update legacy 2016/2018 wording to current 2022 terminology.
Kaynes	Publish current EPR registration/target/return evidence if applicable.

10. Conclusion

The review confirms that Indian e-waste compliance cannot be reduced to one identical checklist for all electronics manufacturers. The legal position depends on the company's role, products, brand ownership, imports, manufacturing activities and whether it qualifies as a Producer under the E-Waste Rules 2022.

Several companies demonstrate mature operational controls. The main cross-company weakness is **public transparency of current regulatory evidence**—especially CPCB EPR registration, targets, certificates and returns.

The most defensible conclusion is:

> **“No confirmed violation was established from the reviewed public evidence. However, multiple requirements are not publicly demonstrated and require company/CPCB-level verification.”**

CPCB's current portal guidance makes clear that applicable Producers must achieve applicable EPR targets and file periodic returns, and the current framework includes environmental-compensation mechanisms for non-compliance. [R11][R14]

Annexure A — Company Evidence Matrix

Legend: ✓ documented; ~ partial/role-dependent; ? not established in reviewed evidence; N/A not applicable based on disclosed role.

Company	EPR / role	Recycler	RoHS	Collection / EOL	Returns / target
Foxconn India	?	~	?	~	?
Dixon	~	✓	✓	~	?
Elin	✓ / current scope to verify	~	?	~	?
VVDN	~	✓	?	✓	?
Syrma SGS	✓ / role-dependent	✓	?	✓ / B2B limit	?
Tata Electronics	~ / role-dependent	~	?	~	?
Avalon	?	?	?	~	?
Centum	N/A / role-dependent	✓	?	~	?
Sahasra	✓	✓	✓	✓	~
Kaynes	~	✓	?	~	?

Documentary highlights

Dixon — authorised recyclers and RoHS controls [R1]. Elin — current EPR/plastic registration and historical e-waste COP [R2]. VVDN — EPR plan and recycler arrangement [R3]. Syrma SGS — authorised recycler and collection points [R4]. Tata Electronics — government import/end-of-life condition [R5]. Avalon — e-waste quantity disclosed [R6]. Centum — e-waste quantity and recovery/disposal data [R7]. Sahasra — EPR/e-waste authorisation and collection [R8]. Kaynes — authorised waste handling [R9].

Annexure B — Regulatory & Documentary Sources

R11 — CPCB E-Waste Management System / EPR Portal: current portal guidance on registration, EPR targets, EPR certificates and quarterly returns.

R12 — CPCB FAQ under E-Waste (Management) Rules, 2022: applicability, EPR, registration, recycling and RoHS requirements.

R13 — MOKO Technology, "Top 10 Electronics Manufacturers in India (2026 updated)": source corresponding to the supplied reference image and correct ten-company set.

R14 — CPCB Environmental Compensation Guidelines: framework for environmental compensation for non-compliance.

R1 — Dixon Technologies: annual-report disclosure on authorised recyclers, 3R waste management and RoHS.

R2 — Elin Electronics: FY2024–25 annual report; earlier annual-report material on e-waste management COP.

R3 — VVDN Technologies: official e-waste management page describing EPR plan and recycler/dismantler arrangement.

R4 — Syrma SGS: official EPR compliance page and FY2024–25 annual report.

R5 — MoEFCC: Expert Committee record concerning Tata Electronics Systems Solutions and used EEE import/end-of-life disposal.

R6 — Avalon Technologies: FY2024–25 annual-report filing with e-waste quantity and waste-management data.

R7 — Centum Electronics: FY2024–25 annual report with e-waste and waste recovery/disposal data.

R8 — Sahasra Electronics: official EPR/e-waste pages, certifications and sustainability material.

R9 — Kaynes Technology: annual-report material describing authorised waste handling.

R10 — MoEFCC used-EEE import record: government condition requiring end-of-life disposal under E-Waste Rules 2022 and HOWM Rules for the relevant import.

Source-control note: This is a dated desk review as of 01 September 2026. For regulatory submission or legal advice, current CPCB/SPCB registration, EPR targets, returns, certificates, inspection records and notices should be independently verified.

Executive Dashboard — One-Page Reference

Dimension	Overall observation
Principal law	E-Waste (Management) Rules, 2022 and amendments
Strongest practice	Authorised recycler/channelisation and collection arrangements
Biggest evidence gap	Current CPCB EPR registration, targets, certificates and returns
Key role distinction	Producer vs B2B EMS manufacturer vs waste generator
Confirmed violations	None established from reviewed evidence
Verification priority	CPCB EPR registration and company-specific legal role
Key recommendation	Publish an auditable, current e-waste compliance register

Final statement

The correct conclusion is not that the ten companies “do not follow the rules.” The evidence supports a more precise conclusion: several companies demonstrate substantial e-waste controls, while some current legal requirements are not publicly demonstrated. Those gaps should be verified rather than treated as proven violations.

— End of Report —

TEAM MEMBERS

E-Waste Compliance & Gap Analysis Report

No.	Team Member	Roll No.
1	Nishant Atram	24102A0066
2	Gaurav Yadav	24102A0056
3	Divesh Jadhav	24102A0068
4	Sagar Rathod	24102A0067

Project: India Electronics Manufacturers — E-Waste Compliance & Gap Analysis

Prepared by: The above team members